Galois Theory Through Cyclotomic Symmetry

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Abstract

This paper explores Galois theory through the natural setting of cyclotomic extensions, where the abstractions of field theory are visually guided by symmetry. We develop the framework needed to study algebraic structures through their automorphisms and the Fundamental Theorem of Galois Theory, which allows us to examine how intermediate fields correspond to subgroups of symmetry. We then bring this perspective to life with cyclotomic extensions, obtained by adjoining roots of unity and study the irreducibility of cyclotomic polynomials. In particular, we show that the field generated by a primitive n-th root of unity carries a Galois group naturally isomorphic to $(\mathbb{Z}/n\mathbb{Z})^\times$, revealing a direct connection between number theory and field automorphisms. Through cyclotomic fields, the paper illustrates how symmetry organizes and clarifies the structure of algebraic objects and provides a concrete entry point into algebraic number theory.

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1 Introduction

1.1 Historical Background

The problem of solving polynomial equations has shaped the development of algebra for centuries. By the sixteenth century, mathematicians had discovered formulas expressing the roots of quadratic, cubic, and quartic equations in terms of radicals, leading naturally to the hope that similar formulas might exist for polynomials of arbitrary degree. This expectation was ultimately overturned in the early nineteenth century, when Niels Abel (1802 – 1829) proved that no general solution by radicals exists for polynomials of degree five or higher. His result, known as the Abel–Ruffini theorem, marked a fundamental turning point in the theory of equations.

Building on this development, Évariste Galois (1811 – 1832) introduced a radically new perspective while still in his teens. Rather than studying equations directly, Galois proposed studying the symmetries among their roots. This idea led to the notion of a Galois group, which captures the algebraic relations among the roots of a polynomial through permutations preserving those relations. Although Galois' work was not fully appreciated during his lifetime, it established the profound connection between field theory and group theory that underpins modern algebra.

1.2 The Idea Behind Cyclotomic Fields

At its core, Galois theory shows that algebraic structure is mirrored by symmetry. This perspective becomes especially vivid in the study of cyclotomic fields, obtained by adjoining roots of unity to the field of rational numbers. These fields form one of the most concrete and computable classes of Galois extensions. Their structure is explicit, and their Galois groups can be described in terms of familiar arithmetic objects, namely the multiplicative groups of integers modulo n. As such, they provide an ideal setting in which we can see the ideas of Galois theory in action.

The construction begins with the roots of the polynomial $x^n - 1$, known as the nth roots of unity. Geometrically, these n complex numbers lie on the unit circle in the complex plane, forming the vertices of a regular n-gon. The primitive roots, which generate all others through their powers, are of particular interest, as they serve as the building blocks for the extension field $\mathbb{Q}(\zeta_n)$. We care about these fields not only for their geometric elegance but also because they sit at the intersection of number theory and algebra. They are the testing ground for the Kronecker-Weber theorem and provided the first major steps toward proving Fermat's Last Theorem.

Central to this study is the realization that any automorphism of $\mathbb{Q}(\zeta_n)$ is determined entirely by how it permutes these primitive roots. This leads to the fundamental isomorphism

$$\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times.$$

This result allows us to translate difficult questions about field automorphisms into simple calculations involving modular arithmetic.

The goal of this paper is to develop the theory of cyclotomic fields $\mathbb{Q}(\zeta_n)$, compute their degrees, and describe their Galois groups. Along the way, we build the necessary background in field theory and use cyclotomic extensions as a guiding example to illustrate the broader principles of Galois theory. We conclude by briefly exploring how these ideas extend to Ernst Kummer's (1810 – 1893) generalization, known as Kummer theory, which studies extensions obtained by adjoining nth roots of elements.

The paper is organized as follows. Section 2 develops the algebraic preliminaries needed to study splitting fields and field automorphisms. Section 3 introduces Galois extensions and establishes the Fundamental Theorem of Galois Theory. In Section 4, we apply this machinery to cyclotomic fields, proving the irreducibility of cyclotomic polynomials and computing their Galois groups explicitly. Finally, Section 5 briefly discusses the significance of these concepts and further directions. Throughout this paper, many of our ideas and techniques are based on Ian Stewart's *Galois Theory* [4], J. S. Milne's *Fields and Galois Theory* [3], John Fraleigh's *A First Course in Abstract Algebra* [2], and Keith Conrad's lecture notes on Cyclotomic Extensions [1].

2 Algebraic Preliminaries

2.1 Fields and Polynomials

We begin by establishing the algebraic framework in which our work will take place. This includes several foundational definitions from field theory, along with key examples that will be used throughout.

Definition 2.1: Field

A **field** is a commutative ring with unity in which every nonzero element is invertible. More specifically, a field is a set F equipped with binary operations addition $(+)$ and multiplication $(\cdot)$ such that:

(a) $(F, +)$ is an abelian group,

(b) $(F^\times, \cdot)$ is an abelian group, where $F^\times = F \backslash \{0\}$,

(c) the distributive properties hold.

A **subfield** $S \subseteq F$ is a subring that is closed under multiplicative inverses.

Example 2.1

Standard examples of fields include $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, and the finite field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$, where p is prime.

Definition 2.2: Characteristic

The **characteristic** of a ring R, denoted $\operatorname{char}(R)$, is the smallest positive integer n such that

$$n \cdot a = 0 \quad \text{for all } a \in R,$$

provided such an integer exists. Otherwise, we write $\operatorname{char}(R) = 0$.

We can equivalently define $\operatorname{char}(R)$ as the additive order of the multiplicative identity 1_R. The characteristic measures how the additive structure of a field interacts with its multiplicative identity, and it plays an important role in distinguishing different types of fields.

Theorem 2.1: Characteristic of a Field

If F is a field, then $\text{char}(F)$ *is either* 0 *or a prime number.*

Proof. Let F be a field, and define a ring homomorphism $\psi : \mathbb{Z} \to F$ by $n \mapsto n \cdot 1_F$, where $n \cdot 1_F$ is the identity 1_F added to itself n times. The kernel of ψ, $\ker(\psi)$, is an ideal of $\mathbb{Z}$, thus it must be of the form $n\mathbb{Z}$ for some integer $n \geq 0$.
If $n = 0$, then $\ker(\psi) = \{0\}$, so F will have characteristic 0. Otherwise, if $n > 0$, then n is the smallest integer such that $n \cdot 1_F = 0$, hence $\text{char}(F) = n$.
Now, suppose $\text{char}(F) = n > 0$. Assume that n is composite, meaning we can write $n = ab$, where $1 < a, b < n$. Then,

$$0 = n \cdot 1_F = (ab) \cdot 1_F = (a \cdot 1_F)(b \cdot 1_F).$$

However, since F is a field, it has no zero divisors, so either $a \cdot 1_F = 0$ or $b \cdot 1_F = 0$, contradicting the minimality of n. Therefore, n must be prime. □

To construct field extensions, we will adjoin roots of polynomials. This motivates our study of a polynomial's factorization behavior over a given field.

Definition 2.3: Reducibility

Let R be a subring of $\mathbb{C}$, and let $p(x) \in R[x]$ be a non-constant polynomial. We say that $p(x)$ is **reducible over** R if it can be written as

$$p(x) = f(x)g(x),$$

where $f, g \in R[x]$ have strictly smaller degree than p. Otherwise, $p(x)$ is **irreducible over** R.

Example 2.2

The polynomial $p(x) = x^2 - 9$ is reducible over $\mathbb{Q}$ since we can write

$$x^2 - 9 = (x - 3)(x + 3).$$

In contrast, $x^2 + 9$ is irreducible over $\mathbb{Q}$. If we suppose that

$$x^2 + 9 = (x + b)(x + d),$$

for some $b, d \in \mathbb{Q}$, then comparing coefficients gives $b + d = 0$ and $bd = 9$, so $b^2 = -9$, a contradiction since $b \in \mathbb{Q}$.

Definition 2.4: Roots and Multiplicity

Let $F \subseteq \mathbb{C}$ be a field, and let $p \in F[x]$ be a polynomial. An element $\alpha \in F$ is a **root** or **zero** of p if $p(\alpha) = 0$. Equivalently, α is a root of p if and only if $(x - \alpha)$ divides $p(x)$ in $F[x]$.

Moreover, we say α has **multiplicity** m if

$$(x-\alpha)^m \mid p(x) \quad \text{but} \quad (x-\alpha)^{m+1} \nmid p(x).$$

We say a root of a polynomial is **simple** if it has multiplicity 1, and it is **multiple** if its multiplicity is greater than one.

Definition 2.5: Field Homomorphisms

Let F and E be fields. A map $\psi : F \to E$ is a **field homomorphism** if

$$\psi(x+y) = \psi(x) + \psi(y) \quad \text{and} \quad \psi(xy) = \psi(x)\psi(y)$$

for all $x, y \in F$, and $\psi(1) = 1$.
An **isomorphism** is a bijective homomorphism. An isomorphism from a field to itself is called an **automorphism**. A homomorphism that is injective but not necessarily surjective is a **monomorphism**.

We now introduce the central objects of this paper, which will be covered more thoroughly in later sections. Roots of unity will be the primary objects used to build cyclotomic extensions.

Definition 2.6: Roots of Unity

Let n be a positive integer. An n**th root of unity** in a field F is an element $\zeta \in F$ such that

$$\zeta^n = 1.$$

That is, the nth roots of unity are the roots of $x^n - 1$.

Example 2.3

Over $\mathbb{Q}$, the polynomial $x^4 - 1$ does not split completely. Its roots are $1, -1, i, -i$, and $i \notin \mathbb{Q}$. To obtain all roots, we must extend $\mathbb{Q}$ to a field containing i, which we will define later as $\mathbb{Q}(i)$.

Example 2.4

In $\mathbb{R}$, the only roots of unity are ± 1. In $\mathbb{C}$, for all $n \geq 1$, there are exactly n distinct nth roots of unity, given by

$$\zeta_n^k = e^{2\pi i k/n}, \quad 0 \leq k \leq n.$$

They form a cyclic group under multiplication of order n.

The concepts introduced in this section will be used repeatedly in what follows, often in combination. To fully understand roots of polynomials such as $x^n - 1$, we will need to construct larger fields containing these roots.

2.2 Field Extensions

We now turn to **field extensions**, which form the central framework for our study of Galois theory. Field extensions allow us to enlarge a given field so that polynomials acquire roots that may not exist in the original field. A key example, which we will study more thoroughly, is that of cyclotomic fields obtained by adjoining roots of unity to $\mathbb{Q}$.

Definition 2.7: Field Extension

Let E and F be fields. We say that E is a **field extension** of F, written by E/F, if E contains F as a subfield.
If K is a subset of E, we denote by $F(K)$ the smallest subfield of E containing both F and K. Equivalently, $F(K)$ is obtained from F by adjoining all elements of K. If K has a single element k, we write $F(k)$, and if $K = \{k_1, k_2, \ldots, k_n\}$, we write

$$F(k_1, k_2, \ldots, k_n).$$

Example 2.5

Consider $\mathbb{Q}(i)$, which we construct by adjoining a root of the polynomial $x^2 + 1$ to $\mathbb{Q}$. Every element of $\mathbb{Q}(i)$ can be written uniquely in the form

$$a + bi \quad \text{with } a, b \in \mathbb{Q}.$$

Note that $\mathbb{Q}(i)$ is not equal to $\mathbb{C}$. It consists only of complex numbers with rational real and imaginary parts. Many familiar numbers, such as $\sqrt{2}$ and π are not contained in $\mathbb{Q}(i)$.

Example 2.6

Let $F = \mathbb{Q}$ and $K = \{i, \sqrt{3}\}$. Then the field extension

$$\mathbb{Q}(i, \sqrt{3})$$

contains both generators and their products. Every element in $F(K) = \mathbb{Q}(i, \sqrt{3})$ can be written uniquely as

$$a + bi + c\sqrt{3} + di\sqrt{3},$$

where $a, b, c, d \in \mathbb{Q}$.

Definition 2.8: Algebraic and Transcendental Numbers

Let $F \subseteq \mathbb{C}$ and $\alpha \in \mathbb{C}$.

- We say α is **algebraic over** F if there exists a non-zero polynomial p over F such that $p(\alpha) = 0$.
- Otherwise, α is **transcendental over** F.

A number is **algebraic** if it is algebraic over $\mathbb{Q}$, and transcendental otherwise.

Example 2.7

The number $\sqrt{2}$ is algebraic over $\mathbb{Q}$ because it satisfies $x^2 - 2 = 0$. Similarly, $\sqrt{\pi}$ is algebraic over $\mathbb{Q}(\pi)$ because

$$(\sqrt{\pi})^2 - \pi = 0.$$

Definition 2.9: Algebraic Extension

An extension E/F is an **algebraic extension** if every element of E is algebraic over F.

Definition 2.10: Minimal Polynomial

Let $F \subseteq \mathbb{C}$ be a field and $\alpha \in \mathbb{C}$ algebraic over F. The **minimal polynomial** of α over F is the unique monic polynomial

$$m_\alpha(x) \in F[x]$$

of least degree such that $m_\alpha(\alpha) = 0$.
Recall that a non-zero polynomial is **monic** if its leading coefficient is 1.

Example 2.8

The minimal polynomial of $\sqrt{2}$ over $\mathbb{Q}$ is $x^2 - 2$.

Lemma 2.1

If α is algebraic over a field F, then its minimal polynomial $m_\alpha(x)$ is irreducible over F, and it divides every polynomial $p(x) \in F[x]$ having α as a zero.

Proof. Suppose $m_\alpha(x)$ is reducible for the sake of contradiction, so we can write

$$m_\alpha(x) = f(x)p(x),$$

where $\deg(f), \deg(p) < \deg(m_\alpha)$. Evaluating at α, we obtain

$$0 = m_\alpha(\alpha) = f(\alpha)p(\alpha),$$

so either $f(\alpha) = 0$ or $p(\alpha) = 0$, contradicting the minimality of m_α. Thus, the minimal polynomial m_α is irreducible.
Now, let $p(x) \in F[x]$ satisfy $p(\alpha) = 0$. By polynomial division, dividing $p(x)$ by $m_\alpha(x)$ gives

$$p(x) = q(x)m_\alpha(x) + r(x),$$

where $\deg(r) < \deg(m_\alpha)$. Since $p(\alpha) = 0$, we have

$$0 = p(\alpha) = m_\alpha(\alpha)q(\alpha) + r(\alpha) = r(\alpha).$$

The only polynomial of degree less than that of the minimal polynomial having α as a root must be the zero polynomial, so $r(x) = 0$. This implies $p(x) = m_\alpha(x)q(x)$, so $m_\alpha(x)$ divides $p(x)$. □

Definition 2.11: Simple Extension

A field extension E/F is called a **simple extension** if $E = F(\alpha)$ for some $\alpha \in E$.

Before proceeding, recall that a **basis** of a vector space is a linearly independent spanning set, and the **dimension** is the number of elements in a basis.

Lemma 2.2

Let $m(x) \in F[x]$. Every polynomial $p(x) \in F[x]$ is congruent modulo $m(x)$ to a unique polynomial of degree less than $\deg(m)$.

Proof. By polynomial division,

$$p(x) = m(x)q(x) + r(x),$$

where $q, r \in F[x]$, and $\deg(r) < \deg(m)$. Then, $p - r = mq$, so $p \equiv r \mod m$.
For uniqueness, suppose $r \equiv s \mod m$, with $\deg(r), \deg(s) < \deg(m)$. Then, $r - s$ is divisible by m, but $\deg(r - s) < \deg(m)$, so $r - s = 0$. Hence, $r = s$. □

Theorem 2.2

Let $F(\alpha)/F$ be a simple algebraic extension, and let $m_\alpha(x)$ be the minimal polynomial of α over F such that $\deg(m_\alpha) = n$. *Then,*

$$\{1, \alpha, \alpha^2, \ldots, \alpha^{n-1}\}$$

is a basis for $F(\alpha)$ as a vector space over F.

Proof. Let $p(x) \in F[x]$. Then, $p(\alpha) \in F(\alpha)$. By Lemma 2.2, there exists a unique polynomial $r(x)$ with $\deg(r) < n$ such that

$$p(x) \equiv r(x) \mod m_\alpha(x).$$

Evaluating at α, we get $p(\alpha) = r(\alpha)$, so every element of $F(\alpha)$ is a linear combination of

$$1, \alpha, \ldots, \alpha^{n-1}.$$

To prove linear independence, suppose

$$a_{n-1}\alpha^{n-1} + \cdots + a_1\alpha + a_0 = 0.$$

Then the polynomial

$$r(x) = a_{n-1}x^{n-1} + \cdots + a_1x + a_0$$

satisfies $r(\alpha) = 0$. By Lemma 2.1, $m_\alpha(x) \mid r(x)$, but $\deg(r) < \deg(m_\alpha)$, so $r = 0$, hence all $a_i = 0$.
Thus, the set is linearly independent and forms a basis. □

Definition 2.12: Degree of an Extension

Let E/F be a field extension. The **degree** of E over F, denoted $[E : F]$, is the dimension of E as a vector space over F.
We say that E/F is a **finite extension** if $[E : F] < \infty$.

Example 2.9

A basis of $\mathbb{C}$ over $\mathbb{R}$ is $\{1, i\}$, so $[\mathbb{C} : \mathbb{R}] = 2$.

The next result is fundamental as it explains how degrees behave in towers of field extensions.

Theorem 2.3: Tower Law

Let $F \subseteq L \subseteq E$ be fields. Then E/F is finite if and only if both E/L and L/F are finite. In this case,

$$[E : F] = [E : L][L : F].$$

Proof. Let $T = \{t_1, \ldots, t_n\}$ be a basis of E as a vector space over L and let $A = \{a_1, \ldots, a_n\}$ be a basis of L over F. We claim that

$$AT = \{a_i t_j \mid 1 \leq i \leq m,\ 1 \leq j \leq n\}$$

is a basis of E over F.
First, we show that AT spans E. Let $b \in E$. Since T is a basis over L, we can write

$$b = \sum_{j=1}^{n} d_j t_j, \quad d_j \in L.$$

Each d_j can be expressed in terms of the basis A as

$$d_j = \sum_{i=1}^{m} c_{ij} a_i, \quad c_{ij} \in F.$$

Substituting, we obtain

$$b = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij}(a_i t_j)$$

for all $b \in E$, so AT spans E.
Next, we show linear independence. Suppose

$$b = \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij}(a_i t_j) = 0.$$

We can rewrite this as

$$\sum_{j=1}^{n} \left(\sum_{i=1}^{m} c_{ij} a_i \right) t_j = 0.$$

Because T is linearly independent over L, each coefficient must vanish, so

$$\sum_{i=1}^{m} c_{ij} a_i = 0 \quad \text{for all } j.$$

Also, A is linearly independent over F, so all $c_{ij} = 0$. So, AT is linearly independent and is a basis of E over F. Hence, $[E : F] = mn = [E : L][L : F]$. □

Example 2.10

We compute $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}]$. First, we have

$$[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$$

since $\{1, \sqrt{2}\}$ is a basis. Moreover, $\sqrt{3} \notin \mathbb{Q}(\sqrt{2})$, and $\{1, \sqrt{3}\}$ is a basis of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}(\sqrt{2})$, so

$$[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})] = 2.$$

By the Tower Law,

$$[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})]\,[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 4.$$

Corollary 2.1: Extended Tower Law

If $F_0 \subseteq F_1 \subseteq \cdots \subseteq F_n$ is a chain of fields, then

$$[F_n : F_0] = \prod_{i=1}^{n} [F_i : F_{i-1}].$$

This can simply be proven by induction on the Tower Law.

Theorem 2.4

Let E/F be an algebraic extension. Then, there exists a finite number of elements $\alpha_1, \ldots, \alpha_n$ in E such that $E = F(\alpha_1, \ldots, \alpha_n)$ if and only if E is a finite-dimensional vector space over F, that is, if and only if E is a finite extension of F.

Proof. ($\Rightarrow$) Assume $E = F(\alpha_1, \ldots, \alpha_n)$. Since E is algebraic over F, we have that each α_i is algebraic over F, thus each α_i is algebraic over every extension of F, where $1 \leq i \leq n$. Therefore, $F(\alpha_i)$ is algebraic over F, and $F(\alpha_i, \ldots, \alpha_j)$ is algebraic over $F(\alpha_i, \ldots, \alpha_{j-1})$, for $2 \leq j \leq n$. Then, we can apply the extended tower law to get

$$[F(\alpha_1, \ldots, \alpha_n) : F] = [F(\alpha_1, \ldots, \alpha_n) : F(\alpha_1, \ldots, \alpha_{n-1})] \ldots [F(\alpha_1) : F],$$

where each degree is finite. Thus, we know $[E : F]$ is finite.
($\Leftarrow$) Assume $[E : F]$ is finite. If $[E : F] = 1$, then $E = F$, and we're done. If $[E : F] > 1$, assume $\alpha_1 \in E$ such that $\alpha_1 \notin F$. Then, because E is algebraic over F, we know that α_1 is algebraic over F

and $[F(\alpha) : F] > 1$. Now, if $F(\alpha_1) = E$, we're done. If not, then there exists an $\alpha_2 \in E, \alpha_2 \notin F$ such that α_2 is also algebraic over F and $[F(\alpha_1, \alpha_2) : F] > 1$. If $F(\alpha_1, \alpha_2) = E$, we're done. If not, we can continue this process by adding α_i until we reach some α_n that is algebraic over F and such that $F(\alpha_1, \ldots, \alpha_n) = E$, which is finite due to the Tower Law, since E is a finite (algebraic) extension of F. □

Theorem 2.5: Algebraic Closure

Let E be an extension field of F. Then, the ***algebraic closure of F in E****,*

$$\overline{F}_E = \{\alpha \in E \mid \alpha \text{ is algebraic over } F\},$$

is a subfield of E.

Proof. Note that $0, 1 \in \overline{F}_E$ as they are roots of the polynomials x and $x - 1$ in $F[x]$, respectively. Let $\alpha, \beta \in \overline{F}_E$. Since α and β are algebraic over F, the extension $F(\alpha, \beta)$ is a finite extension of F (with degree $[F(\alpha, \beta) : F] \leq [F(\alpha) : F][F(\beta) : F]$). Every finite extension is algebraic, so every element of $F(\alpha, \beta)$ is algebraic over F.
Since $\alpha \pm \beta, \alpha\beta$, and α/β (for $\beta \neq 0$) all lie in $F(\alpha, \beta)$, they are necessarily algebraic over F. It follows that these elements lie in $\overline{F}_E$. Therefore, $\overline{F}_E$ is a subfield of E. □

Theorem 2.6

The set of all algebraic numbers forms a field.

Proof. Let

$$\overline{\mathbb{Q}} = \{\alpha \in \mathbb{C} \mid \alpha \text{ is algebraic over } \mathbb{Q}\}.$$

The elements in $\overline{\mathbb{Q}}$ are exactly the algebraic numbers. By applying the previous theorem with $F = \mathbb{Q}$ and $E = \mathbb{C}$, we see that $\overline{\mathbb{Q}}$ is the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$. Consequently, $\overline{\mathbb{Q}}$ is a subfield of $\mathbb{C}$, and thus the algebraic numbers form a field. □

Definition 2.13

A field F is **algebraically closed** if every non-constant polynomial $p(x) \in F[x]$ has a root in F.

Note that a field F may be its own algebraic closure in E without being algebraically closed. For example, $\mathbb{Q}$ is its own algebraic closure in $\mathbb{Q}(x)$, yet $\mathbb{Q}$ is not algebraically closed because $x^2 - 2 \in \mathbb{Q}[x]$ has no roots in $\mathbb{Q}$.

Theorem 2.7

Let $F(\alpha)/F$ be a simple extension. If α is transcendental over F, then $[F(\alpha) : F] = \infty$. If α is algebraic over F with a minimal polynomial of degree n, then $[F(\alpha) : F] = n$.

Proof. If α is transcendental over F, the set $\{1, \alpha, \alpha^2, \ldots\}$ is linearly independent over F, implying an infinite-dimensional vector space. If α is algebraic over F, the result follows from the property that

$\{1, \alpha, \ldots, \alpha^{n-1}\}$ forms a basis for $F(\alpha)$ over F. □

Definition 2.14: Splitting Field

A field E containing F is the **splitting field** of $p(x) \in F[x]$ if $p(x)$ **splits**, or factors into a product of linear factors, in $E[x]$ as

$$p(x) = a\prod_{i=1}^{n}(x - \alpha_i), \quad \alpha_i \in E, a \in F,$$

and $E = F(\alpha_1, \ldots, \alpha_n)$.

Example 2.11

The splitting field of $x^2 - 2$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt{2})$. The splitting field of $x^4 - 5x^2 + 6$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt{2}, \sqrt{3})$.

Theorem 2.8: Finite Subgroups of a Field are Cyclic

The group of nth roots of unity in a field is cyclic. More generally, every finite subgroup of $F^\times$ is cyclic.

Proof. Let F be a field and let $G \subseteq F^\times$ be a finite subgroup. Since multiplication in F is commutative, G is an abelian group. Let $|G| = n$. A standard result from abelian group theory states that if G contains elements of orders n_1 and n_2, then it contains an element of order $\operatorname{lcm}(n_1, n_2)$. Let N be the largest (maximal) order of any element in G; that is,

$$N = \max\{\operatorname{ord}(g) : g \in G\}.$$

We claim that the order of every element of G divides N. Let $g \in G$ have order m, meaning that m is the smallest positive integer such that $g^m = 1$. Then there exists an element of order $\operatorname{lcm}(m, N) \geq N$ in G. By maximality of N, we must have

$$\operatorname{lcm}(m, N) \leq N,$$

hence $\operatorname{lcm}(m, N) = N$. This implies that $m \mid N$. Therefore, every element $g \in G$ satisfies $g^N = 1$, so G is contained in the set of roots of the polynomial $x^N - 1$. Since a polynomial of degree N has at most N roots in a field, it follows that $|G| \leq N$. On the other hand, by Lagrange's theorem, the order of any element in G divides $|G|$, so N divides $|G|$. Hence $N \leq |G|$. We conclude that $N = |G|$, so there exists an element in G of order $|G|$. Therefore, G is cyclic. □

Note that in a field of prime characteristic p, the only pth root of unity is 1. If $x^p = 1$, then

$$x^p - 1 = (x - 1)^p$$

by the binomial theorem in characteristic p. Thus $x^p - 1 = 0$, which implies $x = 1$. Equivalently, the equation $x^p = 1$ has a unique solution in F. This fact will be important when studying roots of unity in fields of positive characteristic. This also reflects the fact that the Frobenius map $x \mapsto x^p$ is injective in any field of characteristic p.

Definition 2.15: Primitive Root of Unity

An nth root of unity is **primitive** if its order is exactly n, That is,

$$\zeta^k \neq 1 \quad \text{for all } 1 \leq k \leq n.$$

Example 2.12

Let α be a root of $p(x) = x^n - a$. The roots of p are

$$\{\alpha, \zeta\alpha, \zeta^2\alpha, \ldots, \zeta^{n-1}\alpha\},$$

where ζ is a primitive n-th root of unity ($\zeta^n = 1$). The splitting field is $F(\alpha)$ if and only if $\zeta \in F$.

Cyclotomic extensions arise as splitting fields of $x^n - 1$, making them natural examples of Galois extensions.

2.3 Separable and Normal Extensions

In order for automorphisms to permute roots meaningfully, polynomials must split into distinct roots. This motivates our focus on the separability and normality of extensions.

Let F be a field of prime characteristic $p \neq 0$, and suppose $a \in F$ is not a pth power. For example, $x^p - a$ is irreducible in $F = \mathbb{F}_p[x]$, but in its splitting field, we have

$$(x^p - a) = (x - a)^p,$$

so an irreducible polynomial may have multiple roots.

Definition 2.16: Derivative

For a polynomial $f(x) = \sum a_i x^i$, define its **derivative** by

$$f'(x) = \sum i a_i x^{i-1}.$$

We assume the usual rules for sums and products. Note that in a field of characteristic p, the derivative of x^p is 0.

Lemma 2.3

A root α of f is multiple if and only if it is also a root of f'.

Proof. If α is a root of f of multiplicity $n \geq 1$, then we can write

$$f(x) = (x - \alpha)^n g(x),$$

where $g(\alpha) \neq 0$. By the product rule for derivatives,

$$f'(x) = \begin{cases} g(x) + (x - \alpha)g'(x), & n = 1, \\ n(x - \alpha)^{n-1} g(x) + (x - \alpha)^n g'(x), & n > 1. \end{cases}$$

Evaluating at $x = \alpha$,

$$f'(\alpha) = \begin{cases} g(\alpha), & n = 1, \\ 0, & n > 1. \end{cases}$$

Since $g(\alpha) \neq 0$, we conclude that $f'(\alpha) = 0$ if and only if $n > 1$. □

Theorem 2.9

Let $f \in F[x]$ be a non-constant irreducible polynomial. The following are equivalent:

(a) f has a multiple root;

(b) $\gcd(f, f') \neq 1$;

(c) F has characteristic $p > 0$ and $f \in F[x^p]$;

(d) all roots of f are multiple.

Proof. We prove the cycle of implications $(a) \Rightarrow (b) \Rightarrow (c) \Rightarrow (d) \Rightarrow (a)$.
$(a) \Rightarrow (b)$: If f has a multiple root α, then by Lemma 2.3, α is also a root of f'. Hence f and f' share the common factor $(x - \alpha)$, so $\gcd(f, f') \neq 1$.
$(b) \Rightarrow (c)$: Since f is irreducible and $\gcd(f, f') \neq 1$, it follows that f divides f'. However, $\deg(f') < \deg(f)$ unless $f' = 0$, so we must have $f' = 0$. Thus, if

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where $n \geq 1$, then we have

$$f'(x) = n a_n x^{n-1} + (n-1) a_{n-1} x^{n-2} + \cdots + a_1.$$

Thus, $f' = 0$ if and only if $i a_i = 0$ for all i. This forces $\operatorname{char}(F) = p > 0$ and $a_i = 0$ whenever $p \nmid i$. Hence, $f(x)$ can be written in the form

$$f(x) = g(x^p)$$

for some $g(x) \in F[x]$.
$(c) \Rightarrow (d)$ Suppose $f(x) = g(x^p)$, and factor g in some splitting field to obtain

$$g(x) = \prod_i (x - \alpha_i)^{m_i}.$$

Then,

$$f(x) = \prod_i (x^p - \alpha_i)^{m_i}.$$

In characteristic p, each α_i admits a pth root in some extension, so we may write $\alpha_i = \beta_i^p$. Hence,

$$x^p - \alpha_i = (x - \beta_i)^p,$$

and therefore,

$$f(x) = \prod_i (x - \beta_i)^{p m_i}.$$

Thus, every root of f has multiplicity divisible by p, so that all roots of f are multiple.
$(d) \Rightarrow (a)$: This is trivial. If every root of f is multiple, then f has a multiple root. □

Theorem 2.10

Let $f \in F[x]$ be a non-zero polynomial. Then, $\gcd(f, f') = 1$ *if and only if f has only simple roots (in a splitting field). If f satisfies the conditions in Theorem 2.9, we say that f is a* ***separable polynomial****.*

Proof. Let E/F be a splitting field of f. By Lemma 2.3, a root $\alpha \in E$ is multiple if and only if it is also a root of f'. If $\gcd(f, f') = 1$, then f and f' have no common factors in E, so all roots of f are simple.
Conversely, if all roots of f are simple, then f and f' do not share any roots, hence no common factors. Thus, $\gcd(f, f') = 1$. □

More simply, we can say that an irreducible polynomial $f \in F[x]$ is separable over F if and only if f has all zeroes of multiplicity 1.

Definition 2.17: Perfect Field

A field F is **perfect** if either $\text{char}(F) = 0$, or $\text{char}(F) = p > 0$ and every element of F is a pth power in F.
By "every element is a pth power", we mean that for all $a \in F$, there exists a $b \in F$ such that $a = b^p$.

To understand Galois groups of cyclotomic fields, we must ensure roots can be permuted independently by automorphisms, thus we study separability.

Definition 2.18: Separable Extension

An algebraic extension E/F is **separable** if the minimal polynomial of every element of E over F is separable. Otherwise, it is **inseparable**.

In other words, E/F is separable if every irreducible polynomial in $F[x]$ that has a root in E has no repeated roots. Moreover, E/F is inseparable if F is not perfect with characteristic $p \neq 0$, and if there exists an element $\alpha \in E$ with minimal polynomial of the form $m(x^p)$, such that $g \in F[x]$.

Example 2.13

The extension $\mathbb{F}_p(\alpha)$ over $\mathbb{F}_p(\alpha^p)$ is inseparable since the minimal polynomial of α over $\mathbb{F}_p(\alpha^p)$ is $x^p - \alpha^p$, which has a zero derivative.

Definition 2.19: Normal Extension

An algebraic extension E/F is **normal** if the minimal polynomial of every element in E splits in $E[x]$. That is, an algebraic extension E/F is normal if every irreducible polynomial in $F[x]$ having at least one root in E splits completely over E.

If E/F is both normal and separable, then every irreducible polynomial in $F[x]$ that has roots in E splits into distinct linear factors in $E[x]$. In particular, such a polynomial of degree n has exactly n distinct

roots in E. Hence, E/F is normal and separable if and only if every irreducible polynomial in $F[x]$ with a root in E has $\deg(f)$ roots in E.

This ensures that field extensions behave in a way analogous to splitting fields of polynomials with distinct roots, allowing automorphisms to permute roots freely.

Example 2.14

Consider the polynomial $x^3 - 7$, which has one real root $\sqrt[3]{7}$ and two non-real roots in $\mathbb{C}$. The algebraic extension $\mathbb{Q}(\sqrt[3]{7})/\mathbb{Q}$ is separable because it has only simple roots, but is not normal since it does not split in $\mathbb{Q}(\sqrt[3]{7})$.

Example 2.15

The algebraic extension $\mathbb{F}_p(\alpha)/\mathbb{F}_p(\alpha^p)$ is normal but not separable because the minimal polynomial of α is not separable.

3 Galois Theory

3.1 Galois Extensions and Groups

Definition 3.1: K-automorphism

Let E/F be a field extension, so that F is a subfield of the subfield E of $\mathbb{C}$. An F-**automorphism** of E is an automorphism α of E such that

$$\alpha(k) = k \quad \text{for all } k \in F \tag{3.1}$$

We say that α **fixes** $k \in F$ if (3.1) holds.

Theorem 3.1: Fixed Field

Let G be a group of automorphisms of a field E. Then, the ***fixed field*** *of G,*

$$E^G = \{\alpha \in E \mid \sigma\alpha = \alpha \text{ for all } \sigma \in G\},$$

is a subfield of E.

Proof. Let $\alpha, \beta \in E^G$, then $\sigma(\alpha) = \alpha$ and $\sigma(\beta) = \beta$ for all $\sigma \in G$. Therefore, we have

$$\sigma(\alpha \pm \beta) = \sigma(\alpha) \pm \sigma(\beta) = \alpha \pm \beta$$

and

$$\sigma(\alpha\beta) = \sigma(\alpha)\sigma(\beta) = \alpha\beta$$

for all $\sigma \in G$. Also, if $\beta \neq 0$, then

$$\sigma(\alpha/\beta) = \sigma(\alpha)/\sigma(\beta) = \alpha/\beta.$$

Because all σ are automorphisms, we have that

$$\sigma(0) = 0 \quad \text{and} \quad \sigma(1) = 1.$$

Then, $0, 1 \in E^G$, and so E^G is a subfield of E. □

Theorem 3.2

If E/F is a field extension, then the set of all F-automorphisms of E forms a group under composition of maps which we denote by $\mathrm{Aut}(E/F)$.

Proof. Let σ and τ be F-automorphisms of E, then $\sigma\tau$ is also an automorphism. Further, if $k \in F$, then $\sigma\tau(k) = \sigma(k) = k$, so $\sigma\tau$ is an F-automorphism. The identity map on E is then an F-automorphism. Finally, σ^{-1} is an automorphism of E, and for any $k \in F$, we have

$$k = \sigma^{-1}\sigma(k) = \sigma^{-1}(k)$$

so that σ^{-1} is an F-automorphism. Composition of maps is associative, so the set of all F-automorphisms of E forms a group. □

Definition 3.2: Galois Group

A field extension E/F is **Galois** if it is finite, normal, and separable. Moreover, the **Galois group** $\mathrm{Gal}(E/F)$ of E over F is $\mathrm{Aut}(E/F)$, the group of all F-automorphisms of E.

Example 3.1

Consider the extension $\mathbb{C}/\mathbb{R}$, and suppose that σ is an $\mathbb{R}$-automorphism of $\mathbb{C}$. Let $j = \sigma(i)$, where $i = \sqrt{-1}$. Then, $j^2 = (\sigma(i))^2 = \sigma(i^2) = \sigma(-1) = -1$ since $\sigma(r) = r$ for all $r \in \mathbb{R}$. Then, we have that $j = i$ or $j = -i$.

For any $x, y \in \mathbb{R}$, we have that $\sigma(x + iy) = \sigma(x) + \sigma(iy) = \sigma(x) + \sigma(i)\sigma(y) = x + jy$. So, the two $\mathbb{R}$-automorphisms are

$$\sigma_1 : x + iy \mapsto x + iy$$
$$\sigma_2 : x + iy \mapsto x - iy$$

Notice that σ_1 is the identity, so it is an $\mathbb{R}$-automorphism of $\mathbb{C}$. The map σ_2 is complex conjugation and is also an automorphism. Moreover, $\sigma_2(x + 0i) = x - 0i = x$, so σ_2 is an $\mathbb{R}$-automorphism. Since $\sigma_2^2 = \sigma_1$, the Galois group $\mathrm{Gal}(\mathbb{C}/\mathbb{R})$ is a cyclic group of order 2.

Theorem 3.3

Consider the extension E/F. The following statements are equivalent:

(a) There exists a separable polynomial $p \in F[x]$ whose splitting field is E;

(b) E/F is finite, and the fixed field of all automorphisms of E that fix F is exactly F; that is, $F = E^{\mathrm{Aut}(E/F)}$;

(c) *There exists a finite group G of automorphisms of E such that F is precisely the set of elements fixed by every element in G, i.e., $F = E^G$;*

(d) *The extension E/F is Galois.*

Proof. We will prove each implication sequentially.
(a)⇒(b) Since E is the splitting field of a separable polynomial over F, we have that E is finite over F. Also, p being separable means that all its roots are distinct, and automorphisms of E permute those roots freely. If we let $E^{\mathrm{Aut}(E/F)}$ be the fixed field, we have that

$$[E : E^{\mathrm{Aut}(E/F)}] = |\mathrm{Aut}(E/F)|.$$

But we also know that $|\mathrm{Aut}(E/F)| = [E : F]$, so

$$[E : E^{\mathrm{Aut}(E/F)}] = [E : F].$$

This then forces $E^{\mathrm{Aut}(E/F)} = F$.
(b)⇒(c) Take $G = \mathrm{Aut}(E/F)$. Then, G is finite since E/F is finite, and, by assumption, its fixed field is exactly F. Hence, $F = E^G$.
(c)⇒(d) Assume there exists a finite group $G \subseteq \mathrm{Aut}(E)$ such that $F = E^G$. To show E/F is Galois, we need to show that it is normal and separable. First, to show separability, let $\alpha \in E$, and consider

$$\prod_{\sigma \in G} (x - \sigma(\alpha)).$$

This polynomial lies in $F[x]$ because it is fixed by all $\sigma \in G$, and its roots are distinct. Therefore, α is separable over F, so E/F is separable.
Now, notice that this same polynomial shows that all conjugates of α under G lie in E. Thus, every irreducible polynomial over F that has one root in E splits completely in E, so E/F is normal. Therefore, E/F is Galois.
(d)⇒(a) Assume that E/F is Galois. We want a separable polynomial whose splitting field is E. First, since E/F is finite, we can write

$$E = F(\alpha_1, \ldots, \alpha_n),$$

where $\alpha_1, \ldots, \alpha_n$ are generators. Let $p_i(x)$ be the minimal polynomial of α_i over F, thus each p_i is separable. Now, we can consider

$$p = p_1(x) \cdots p_n(x),$$

which is also separable. Since each $\alpha_i \in E$, we conclude by normality that E contains all roots of p_i. Therefore, E contains all roots of p, or is generated by these roots, and so E is the splitting field of p over F. □

Definition 3.3

If a polynomial $p \in K[x]$ is separable, then its splitting field K_p is Galois over K, thus we define $\mathrm{Gal}(K_p/K)$ as the **Galois group** $\mathrm{Gal}(p)$ of p.

Consider the polynomial $p(x) = \prod_{i=1}^{n}(x - \alpha_i)$ in a splitting field K_p. The elements of the Galois group $\mathrm{Gal}(K_p/K)$ map the roots of p to other roots of p, thus they are automorphisms that act as permutations on the set of roots $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$. The splitting field K_p is generated by the α_i, so we can uniquely define the group $\mathrm{Gal}(K_p/K)$ by its permutation.

Therefore, $\mathrm{Gal}(p)$ is mapped injectively to a subset of the symmetric group

$$\mathrm{Sym}(\{\alpha_1, \alpha_, \ldots, \alpha_n\}) \cong \mathbb{S}_n$$

through the action $\sigma(\alpha_i) = \alpha_{\sigma(i)}$. Moreover, the image consists precisely of those permutations preserving all polynomial relations over K among the roots. In particular, $\mathrm{Gal}(p)$ consists of the permutations σ of the set of roots $\{\alpha_1, \alpha_2, \ldots, \alpha_n\}$ such that if $f \in K[x_1, \ldots, x_n]$ satisfies $f(\alpha_1, \ldots, \alpha_n) = 0$, then

$$f(\sigma(\alpha_1), \sigma(\alpha_2), \ldots, \sigma(\alpha_n)) = 0.$$

3.2 The Fundamental Theorem of Galois Theory

Definition 3.4: Intermediate Field

If E/F is a field extension, then a field M such that $F \subseteq M \subseteq E$ is an **intermediate field**. To each intermediate field M, we associate the group $\mathrm{Gal}(E/M)$ of all M-automorphisms of E.

Now, we are finally ready to tackle one of the most important theorems of Galois theory. Note that we will only work with finite extensions here because the infinite case requires concepts from topology which are beyond the scope of this article.

Theorem 3.4: Fundamental Theorem of Galois Theory

Let E/F be a finite Galois extension, and let $G = \mathrm{Gal}(E/F)$ be its Galois group. Then, there is a one-to-one correspondence between the subgroups H of G and the intermediate fields between E and F:

$$\{\textit{subgroups } H \leq G\} \longleftrightarrow \{\textit{intermediate fields } F \subseteq K \subseteq E\}$$

given by

$$H \mapsto E^H = \{x \in E \mid \sigma(x) = x \textit{ for all } \sigma \in H\}$$

and

$$K \mapsto \mathrm{Gal}(E/K) = \{\sigma \in G \mid \sigma(x) = x \textit{ for all } x \in K\}.$$

This correspondence satisfies the following properties:

(a) If $H_1 \subseteq H_2$, then $E^{H_2} \subseteq E^{H_1}$.

(b) The maps are inverses. That is, $\mathrm{Gal}(E/E^H) = H$ for any subgroup $H \leq G$, and $E^{\mathrm{Gal}(E/K)} = K$ for any intermediate field K.

(c) Degree-Order Relationship: $[E : E^H] = |H|$ and $[E : K] = |\mathrm{Gal}(E/K)|$.

(d) Conjugation Compatibility: $E^{\sigma H \sigma^{-1}} = \sigma(E^H)$ and $\mathrm{Gal}(E/\sigma K) = \sigma\,\mathrm{Gal}(E/K)\sigma^{-1}$ for all $\sigma \in G$.

(e) Normality Correspondence: K/F is Galois if and only if $\mathrm{Gal}(E/K) \trianglelefteq G$.

Proof. Let $H \leq G$. Define

$$K = E^H = \{x \in E \mid \sigma(x) = x \text{ for all } \sigma \in H\}.$$

We first check that K is a field. If $x, y \in K$, then for any $\sigma \in H$,

$$\sigma(x+y) = \sigma(x) + \sigma(y), \quad \sigma(xy) = \sigma(x)\sigma(y) = xy, \quad \sigma(x^{-1}) = (\sigma(x))^{-1} = x^{-1} \text{ (if } x \neq 0).$$

Thus, K is closed under addition, multiplication, and inverses, so K is a field. Also, $F \subseteq K$ because every automorphism in G fixes F. Therefore, K is an intermediate field.
Similarly, for any intermediate field K, the set

$$\mathrm{Gal}(E/K) = \{\sigma \in G \mid \sigma(x) = x \text{ for all } x \in K\}$$

is a subgroup of G because the identity fixes K, composition preserves fixing K, and inverses also fix K.

(a) If $H_1 \subseteq H_2$, then any x fixed by all $\sigma \in H_2$ is certainly fixed by all $\sigma \in H_1$. Hence, $E^{H_2} \subseteq E^{H_1}$.

(b) For $H \leq G$, clearly $H \subseteq \mathrm{Gal}(E/E^H)$ since every $\sigma \in H$ fixes E^H. But the degree formula gives

$$[E : E^H] = |H| \quad \text{and} \quad |\mathrm{Gal}(E/E^H)| \leq [E : E^H] = |H|.$$

Therefore, $|H| \leq |\mathrm{Gal}(E/E^H)| \leq |H|$, so equality holds, and $\mathrm{Gal}(E/E^H) = H$.

Now, for any intermediate field K, let $H = \mathrm{Gal}(E/K)$. Then $[E : E^H] = |H| = [E : K]$, and clearly $K \subseteq E^H$. Since the degrees match, we must have $K = E^H$.

(c) For $H \leq G$, we have $[E : E^H] = |H|$. More generally, if $H_1 \subseteq H_2$, then

$$[E^{H_1} : E^{H_2}] = \frac{[E : E^{H_2}]}{[E : E^{H_1}]} = \frac{|H_2|}{|H_1|} = (H_2 : H_1),$$

which is the index of H_1 in H_2.

(d) For any $\sigma \in G$,

$$x \in E^{\sigma H \sigma^{-1}} \iff \tau(x) = x \text{ for all } \tau \in \sigma H \sigma^{-1} \iff \sigma^{-1}(x) \in E^H \iff x \in \sigma(E^H).$$

Similarly, $\mathrm{Gal}(E/\sigma(K)) = \sigma\,\mathrm{Gal}(E/K)\sigma^{-1}$.

(e) ($\Rightarrow$) If K/F is Galois, then $\sigma(K) = K$ for all $\sigma \in G$. By part (d), this implies $\mathrm{Gal}(E/K)$ is invariant under conjugation in G, i.e., normal.
($\Leftarrow$) Conversely, if $\mathrm{Gal}(E/K) \trianglelefteq G$, then for any $\sigma \in G$, $\sigma\,\mathrm{Gal}(E/K)\sigma^{-1} = \mathrm{Gal}(E/K)$. By injectivity of the correspondence, this implies $\sigma(K) = K$. Hence, K is fixed by all automorphisms and K/F is Galois.

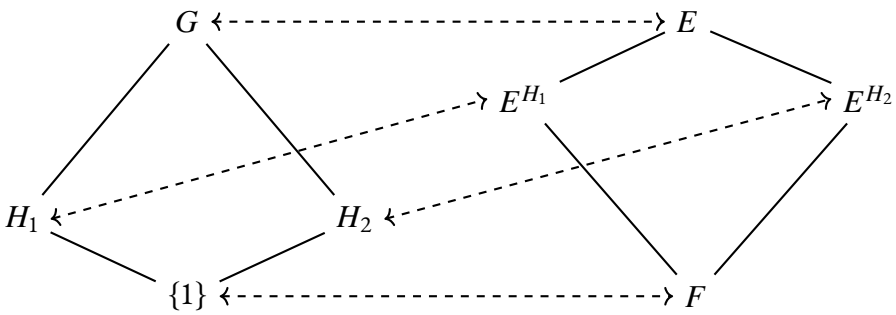


Figure 1: The Galois correspondence

The subgroups of the Galois group G correspond bijectively to intermediate fields of E/F, with larger subgroups corresponding to smaller fields. Solid lines show the subgroup and field lattices, while the horizontal dashed arrows indicate the one-to-one correspondence $H \mapsto E^H$. □

Example 3.2

Consider the field extension $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$. First let $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ and recall from Example 2.10 that $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 4$, so $[K : \mathbb{Q}] = 4$. A field automorphism in $\mathrm{Gal}(K/\mathbb{Q})$ must fix all rational numbers, but it can change the signs of $\sqrt{2}$ and $\sqrt{3}$. Then, because $(\pm\sqrt{2})^2 = 2$ and $(\pm\sqrt{3})^2 = 3$, each square root may independently be sent to either its positive or negative value. Thus there are four automorphisms:

$$
\begin{aligned}
e &: \sqrt{2} \mapsto \sqrt{2}, \quad \sqrt{3} \mapsto \sqrt{3};\\
\sigma &: \sqrt{2} \mapsto -\sqrt{2}, \quad \sqrt{3} \mapsto \sqrt{3};\\
\tau &: \sqrt{2} \mapsto \sqrt{2}, \quad \sqrt{3} \mapsto -\sqrt{3};\\
\sigma\tau &: \sqrt{2} \mapsto -\sqrt{2}, \quad \sqrt{3} \mapsto -\sqrt{3}.
\end{aligned}
$$

The group operation is composition of automorphisms. Each nontrivial element has order 2, and the group is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, the Klein four group. Now, by Theorem 3.4, since the Galois group has 5 subgroups, there must be five intermediate fields, which we will now find.
First, the entire Galois group fixes only the rational numbers, so $K^{\mathrm{Gal}(K/\mathbb{Q})} = \mathbb{Q}$. Also, for the subgroup $\langle\sigma\rangle$, the automorphism changes the sign of $\sqrt{2}$ but leaves $\sqrt{3}$ unchanged, so the fixed field is $K^{\langle\sigma\rangle} = \mathbb{Q}(\sqrt{3})$. Similarly, the subgroup $\langle\tau\rangle$ fixes $\sqrt{2}$ and changes the sign of $\sqrt{3}$, so $K^{\langle\tau\rangle} = \mathbb{Q}(\sqrt{2})$. As for the subgroup $\langle\sigma\tau\rangle$, since it changes the signs of both square roots but keeps the sign of the product $\sqrt{6}$ fixed, the fixed field is $K^{\langle\sigma\tau\rangle} = \mathbb{Q}(\sqrt{6})$. Finally, the trivial subgroup e fixed every element in K, so its fixed field is simply $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Hence we've shown the correspondence between the subgroups of the Galois group and the intermediate fields.

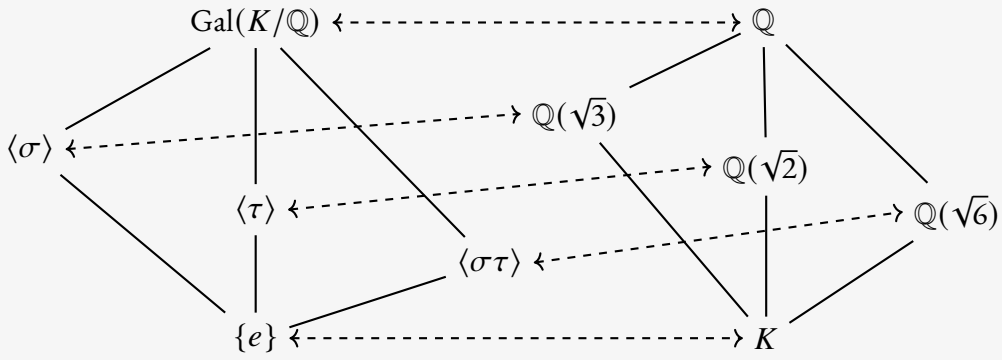


Figure 2: The Galois Correspondence in Example 3.2

4 Cyclotomic Extensions

4.1 Roots of Unity and Cyclotomic Polynomials

Recall from Section 2.1 that we defined an nth root of unity to be a root of the polynomial $x^n - 1$. In Section 2.2, we introduced the primitive nth root of unity as an nth root of unity of order n. We now provide a more formal definition.

Definition 4.1

A **primitive** nth **root of unity** in a field F is an element of order n in the multiplicative group $F^\times$. Such an element only exists if $\text{char}(F) \nmid n$.

Example 4.1

In $\mathbb{C}$, the number -1 is a fourth root of unity because $(-1)^4 = 1$, but it is not a *primitive* fourth root of unity because it has order 2 since $(-1)^2 = 1$.

Definition 4.2: Euler's Totient Function

Denoted by $\varphi(n)$, **Euler's totient function** counts the number of positive integers less than or equal to n that are relatively prime to n. It gives the order of multiplicative group of integers modulo n, denoted $(\mathbb{Z}/n\mathbb{Z})^\times$. That is,

$$\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|.$$

Moreover, we can use **Euler's product formula**

$$\varphi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$$

where the product is taken over the distinct primes p dividing n.

Example 4.2

The integers less than or equal to 6 that are relatively prime to 6 are 1 and 5, so

$$\varphi(6) = |(\mathbb{Z}/6\mathbb{Z})^\times| = 2.$$

Lemma 4.1

Let k be a positive integer and a be an element of order n in a group G. Then, the order of a^k in G is

$$\operatorname{ord}(a^k) = \frac{n}{\gcd(n,k)}.$$

Proof. Let $d = \gcd(n,k)$. We seek the smallest positive integer m such that $(a^k)^m = 1$. Since a has order n, this occurs if and only if $n \mid km$. Writing $n = dn'$ and $k = dk'$, the condition $dn' \mid dk'm$ reduces to $n' \mid k'm$. But $\gcd(n', k') = 1$, so n' must divide m. The smallest such positive integer is $m = n' = n/d$, concluding the proof. □

Theorem 4.1

Let ζ_n be a primitive nth root of unity. Then ζ_n^a is a primitive nth root of unity if and only if $\gcd(a,n) = 1$.

Proof. This follows directly from Lemma 4.1. The order of ζ_n^a is $n/\gcd(n,a)$, which equals n if and only if $\gcd(n,a) = 1$. □

This theorem implies that if a field contains all n distinct nth roots of unity, exactly $\varphi(n)$ of them are primitive.

Definition 4.3

For a field F, an extension of the form $F(\zeta)$, where ζ is an nth root of unity, is called a **cyclotomic extension** of F.

Throughout this section, we assume n is the order of the root of unity ζ in $F(\zeta)/F$. Hence, if $x^n - 1$ splits over F, it has n distinct roots which form a finite subgroup $\mu_n \subseteq F^\times$. Since every finite subgroup of a field's multiplicative group is cyclic, μ_n is cyclic and generated by a primitive nth root of unity. Therefore, if $\operatorname{char}(F) \nmid n$, the polynomial $x^n - 1$ is separable.

Let μ_n denote the group of the n nth roots of unity. Moreover, if ζ_n is a root of unity of order n, then ζ_n generates μ_n.

Example 4.3

In $\mathbb{C}$, the second roots of unity are 1 and -1 so $\mu_2 = \{1, -1\}$, and the fourth roots of unity are 1, -1, i, and $-i$, so we write $\mu_4 = \{1, -1, i, -i\}$.

Lemma 4.2

For $\sigma \in \mathrm{Gal}(F(\mu_n)/F)$, there exists an integer a relatively prime to n, such that $\sigma(\zeta) = \zeta^a$ for all $\zeta \in \mu_n$.

Proof. Let $\sigma \in \mathrm{Gal}(F(\mu_n)/F)$. Since σ is an automorphism and $\sigma(\zeta)^n = \sigma(\zeta^n) = 1$, then σ permutes the elements of μ_n. Let ζ_n be a generator of μ_n. Because automorphisms preserve element order, $\sigma(\zeta_n)$ must also have order n. By Theorem 4.1, $\sigma(\zeta_n) = \zeta_n^a$ for some a with $\gcd(a, n) = 1$. For any $\zeta \in \mu_n$, we have $\zeta = \zeta_n^k$ for some k, so

$$\sigma(\zeta) = \sigma(\zeta_n^k) = \sigma(\zeta_n)^k = (\zeta_n^a)^k = (\zeta_n^k)^a = \zeta^a.$$

Therefore, for every $\zeta \in \mu_n$, $\sigma(\zeta) = \zeta^a$, where a is relatively prime to n. □

Example 4.4

Let $F = \mathbb{Q}$ and $n = 5$. The roots of $x^5 - 1$ are $\mu_5 = \{1, \zeta_5, \zeta_5^2, \zeta_5^3, \zeta_5^4\}$. The Galois group $\mathrm{Gal}(\mathbb{Q}(\mu_5)/\mathbb{Q})$ contains an automorphism σ_2 defined by $\sigma_2(\zeta) = \zeta^2$. Note that σ_2 is a power map only on elements of μ_5. For general $\alpha \in \mathbb{Q}(\mu_5)$, $\sigma_2(\alpha)$ is determined by the field's additive and multiplicative structure. For instance,

$$\sigma_2(3\zeta_5^2 - 7\zeta_5 + 1) = 3(\zeta_5^2)^2 - 7(\zeta_5)^2 + 1 = 3\zeta_5^4 - 7\zeta_5^2 + 1.$$

Theorem 4.2

Let F be a field with $\mathrm{char}(F) \nmid n$, and let E be the splitting field of $x^n - 1$ over F. Then,

(a) E contains a primitive nth root of unity.

(b) If ζ is a primitive nth root of unity in E, then $E = F(\zeta)$.

(c) The extension E/F is Galois, and there is an injective homomorphism

$$\psi : \mathrm{Gal}(E/F) \to (\mathbb{Z}/n\mathbb{Z})^\times, \quad \sigma \mapsto [i]$$

where $\sigma(\zeta) = \zeta^i$.

Proof. We'll tackle each part separately.

(a) Since $\mathrm{char}(F) \nmid n$, the derivative $(x^n - 1)' = nx^{n-1} \neq 0$, so $x^n - 1$ has no repeated roots, so it has exactly n distinct roots in its splitting field E. These roots for a finite subgroup $\mu_n \subseteq E^\times$. By Theorem 2.8, every finite subgroup of a field is cyclic. Therefore, μ_n is cyclic of order n, thus it contains an element of order n. Hence E contains a primitive nth root of unity.

(b) Let ζ be a primitive nth root of unity. Since μ_n is cyclic generated by ζ, every nth root of unity is a power of ζ. Therefore, all roots of $x^n - 1$ lie in $F(\zeta)$. Since E is the splitting field of $x^n - 1$ over F, it follows that $E = F(\zeta)$.

(c) Since E is the splitting field of the separable polynomial $x^n - 1$, the extension E/F is Galois. Let $\sigma \in \mathrm{Gal}(E/F)$. Then,

$$\sigma(\zeta)^n = \sigma(\zeta^n) = 1,$$

so $\sigma(\zeta)$ is also an nth root of unity of order n, and $\sigma(\zeta)$ is a primitive nth root of unity. Since $\mu_n = \langle\zeta\rangle$, every primitive nth root of unity is of the form ζ^i for some $[i] \in (\mathbb{Z}/n\mathbb{Z})^\times$. Thus, there exists a unique class $[i] \in (\mathbb{Z}/n\mathbb{Z})^\times$ such that $\sigma(\zeta) = \zeta^i$.

Define

$$\psi : \mathrm{Gal}(E/F) \to (\mathbb{Z}/n\mathbb{Z})^\times, \quad \psi(\sigma) = [i]$$

If $\psi(\sigma) = [i]$ and $\psi(\tau) = [j]$, then $\tau(\zeta) = \zeta^j$. Therefore,

$$(\sigma\tau)(\zeta) = \sigma(\zeta^j) = \sigma(\zeta)^j = (\zeta^i)^j = \zeta^{ij}.$$

Hence $\psi(\sigma\tau) = [ij] = \psi(\sigma)\psi(\tau)$, so ψ is a homomorphism. To show injectivity, suppose $\psi(\sigma) = [1]$. Then, $\sigma(\zeta) = \zeta$. Since $E = F(\zeta)$, every automorphism σ fixes every element of E, so σ is the identity and ψ is injective.

This completes the proof. □

Remark 4.1

The map ψ need not be surjective. For $F = \mathbb{C}$, in which every polynomial already splits over $\mathbb{C}$, so the Galois group is trivial.
Similarly, when $F = \mathbb{R}$, the only possible non-trivial automorphism is complex conjugation. Thus, since $\overline{\zeta} = \zeta^{-1}$, the image is at most $\{1, -1\}$.
However, for $F = \mathbb{Q}$, the situation is much richer. If $n = p$ is prime, then $[\mathbb{Q}(\zeta_p) : \mathbb{Q}] = p - 1$. Since $|(\mathbb{Z}/p\mathbb{Z})^\times| = p - 1$, the injective homomorphism $\mathrm{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q}) \to (\mathbb{Z}/p\mathbb{Z})^\times$ must also be surjective, and therefore must be an isomorphism.

Definition 4.4: Cyclotomic Polynomial

Let $x^n - 1$ be a polynomial in $\mathbb{Q}[x]$. If we remove all factors $x^d - 1$ where $d < n$ and d divides n from $x^n - 1$, the polynomial that remains is called the **cyclotomic polynomial** Φ_n. Therefore,

$$\Phi_n(x) = \prod_{\substack{1 \le k \le n \\ \gcd(k,n)=1}} (x - e^{2\pi i k/n})$$

is the product over the primitive nth roots of unity.
In other words, all primitive nth roots of unity have a common minimal polynomial in $\mathbb{Q}[x]$, which we call the nth cyclotomic polynomial $\Phi_n(x)$.

Recall that every nth root of unity is a primitive dth root of unity for exactly one positive d such that $d|n$. Therefore, we can write

$$x^n - 1 = \prod_{d|n} \Phi_d(x).$$

These are the first ten cyclotomic polynomials:

$$\begin{aligned}
\Phi_1(x) &= x-1 & \Phi_6(x) &= x^2-x+1\\
\Phi_2(x) &= x+1 & \Phi_7(x) &= x^6+x^5+x^4+x^3+x^2+x+1\\
\Phi_3(x) &= x^2+x+1 & \Phi_8(x) &= x^4+1\\
\Phi_4(x) &= x^2+1 & \Phi_9(x) &= x^6+x^3+1\\
\Phi_5(x) &= x^4+x^3+x^2+x+1 & \Phi_{10}(x) &= x^4-x^3+x^2-x+1
\end{aligned}$$

Example 4.5

We have $\Phi_1(x) = x-1$, $\Phi_2(x) = x+1$, $\Phi_3(x) = x^2+x+1$, and since 1, 2, and 3 are exactly the divisors of 6, we can write

$$\Phi_6(x) = \frac{x^6-1}{(x-1)(x+1)(x^2+x+1)} = x^2-x+1.$$

We can also make use of some properties of cyclotomic polynomials that give rise to useful relations.

(i) $x^n - 1 = \prod_{d|n} \Phi_d(x)$.

(ii) $\Phi_p(x) = x^{p-1} + x^{p-2} + \cdots + x + 1$ for a prime p.

(iii) $\Phi_{p^r}(x) = \Phi_p(x^{p^{r-1}})$.

(iv) $\Phi_{2n}(x) = \Phi_n(-x)$ if n is odd.

(v) $\Phi_{mn}(x) = \Phi_n(x^m)$ if all prime factors of m divide n.

(vi) $\Phi_{p^m r}(x) = \Phi_r(x^{p^m})/\Phi_r(x^{p^{m-1}})$ if p and $m \in \mathbb{N}$ are relatively prime.

Theorem 4.3

The cyclotomic polynomials $\Phi_n(x)$ lie in $\mathbb{Z}[x]$ for $n \geq 1$.

Proof. We proceed by induction. For $n = 1$, $\Phi_1(x) = x - 1 \in \mathbb{Z}[x]$. Now, assume $\Phi_d(x) \in \mathbb{Z}[x]$ for all $d < n$. Then, by definition,

$$\Phi_n(x) = \frac{x^n-1}{\prod\limits_{\substack{d|n\\ d<n}} \Phi_d(x)}.$$

By the inductive hypothesis, the denominator is a polynomial in $\mathbb{Z}[x]$, and $x^n - 1 \in \mathbb{Z}$. Then, since the denominator is monic and divides $x^n - 1$ in $\mathbb{Q}[x]$, this implies that the quotient also lies in $\mathbb{Z}[x]$. Therefore every cyclotomic polynomial has integer coefficients.
In other words, $x^n - 1$ is a monic polynomial and has coefficients in $\mathbb{Z}$, so every monic factor of it in $\mathbb{Q}[x]$ will have coefficients in $\mathbb{Z}$. Therefore, we can say that the cyclotomic polynomials lie in $\mathbb{Z}[x]$. □

Lemma 4.3

The cyclotomic polynomial Φ_n has degree $\varphi(n)$ equal to the order of $(\mathbb{Z}/n\mathbb{Z})^\times$.

Proof. By definition, we write the cyclotomic polynomial as

$$\Phi_n(x) = \prod_{\substack{1\le k\le n \\ \gcd(k,n)=1}} (x - \zeta_n^k),$$

where ζ_n is a primitive nth root of unity. By Theorem 4.1, the primitive nth roots of unity are precisely the powers ζ_n^k such that $\gcd(k, n) = 1$. Therefore, the number of primitive nth roots of unity is equal to the number of integers $k \in [0, n]$ satisfying $\gcd(k, n) = 1$, which is exactly $\varphi(n)$. Therefore, $\deg \Phi_n(x) = \varphi(n)$. □

4.2 Irreducibility of Cyclotomic Polynomials

Lemma 4.4

Let F be a field of characteristic not dividing n, and let ζ be a primitive nth root of unity in some extension of F. The following statements are equivalent:

(a) The nth cyclotomic polynomial Φ_n is irreducible over F;

(b) The degree $[F(\zeta) : F] = \varphi(n)$;

(c) The natural homomorphism $\mathrm{Gal}(F(\zeta)/F) \to (\mathbb{Z}/n\mathbb{Z})^\times$ is an isomorphism.

Proof. By definition, ζ is a root of $\Phi_n(x)$, so the minimal polynomial of ζ over F must divide $\Phi_n(x)$. Since $\deg(\Phi_n(x)) = \varphi(n)$, it follows that $\Phi_n(x)$ is the minimal polynomial if and only if $[F(\zeta) : F] = \varphi(n)$, proving (a) and (b) are equivalent.
Furthermore, the Galois group $\mathrm{Gal}(F(\zeta)/F)$ embeds into $(\mathbb{Z}/n\mathbb{Z})^\times$ via a map $\sigma \mapsto a$, where $\sigma(\zeta) = \zeta^a$. This injection is an isomorphism if and only if $|\mathrm{Gal}(F(\zeta)/F)| = \varphi(n)$. Since

$$[F(\zeta) : F] = |\mathrm{Gal}(F(\zeta)/F)|$$

for Galois extensions, this establishes the equivalence of (b) and (c). □

By this lemma, the study of cyclotomic extensions reduces to the question of the irreducibility of $\Phi_n(x)$. We now establish this landmark result for the rational field.

Theorem 4.4

The nth cyclotomic polynomial $\Phi_n(x)$ is irreducible in $\mathbb{Q}[x]$.

Proof. Let ζ be a primitive nth root of unity and let $f(x) \in \mathbb{Z}[x]$ be its minimal polynomial over $\mathbb{Q}$. Since $\Phi_n(\zeta) = 0$, we have $f(x) \mid \Phi_n(x)$ in $\mathbb{Z}[x]$. It suffices to show that every primitive nth root of unity is also a root of $f(x)$.

Any primitive nth root of unity can be written as ζ^k where $\gcd(k, n) = 1$. By decomposing k into its prime factors, it is enough to show that if α is a root of $f(x)$, then α^p is also a root for any prime $p \nmid n$. Suppose, for a contradiction, that $f(\alpha) = 0$ but $f(\alpha^p) \neq 0$. Since α^p is a root of $\Phi_n(x)$, it must be a root of some other irreducible factor $g(x) \in \mathbb{Z}[x]$ such that $\Phi_n(x) = f(x)g(x)$.
Note that α is a root of $g(x^p)$, implying $f(x) \mid g(x^p)$. Let $\overline{f}$ denote the reduction of f modulo p. Reducing coefficients modulo p, we get

$$g(x^p) \equiv g(x)^p \pmod{p}.$$

Thus, $\overline{f}(x) \mid \overline{g}(x)^p$ in $\mathbb{F}_p[x]$. If $\overline{h}(x)$ is an irreducible factor of $\overline{f}(x)$ in $\mathbb{F}_p[x]$, then $\overline{h}(x)$ must also divide $\overline{g}(x)$. This implies that $\overline{\Phi}_n(x)$ has a repeated factor $\overline{h}(x)$, so it has a multiple root in its splitting field over $\mathbb{F}_p$. However, $\Phi_n(x)$ divides $x^n - 1$, and $x^n - 1$ has no multiple roots in characteristic p when $p \nmid n$ (since its derivative nx^{n-1} is non-zero). This contradiction proves that $f(x) = \Phi_n(x)$. □

Corollary 4.1

For a primitive nth root of unity ζ_n, we have $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$ and

$$\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times.$$

Proof. By Theorem 4.4, $\Phi_n(x)$ is the minimal polynomial of ζ_n over $\mathbb{Q}$. Since $\deg \Phi_n(x) = \varphi(n)$, the degree of the extension is $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$. The extension $\mathbb{Q}(\zeta_n)/\mathbb{Q}$ is the splitting field of $\Phi_n(x)$, making it a Galois extension. By Lemma 4.4, the natural embedding of the Galois group into the group of units $(\mathbb{Z}/n\mathbb{Z})^\times$ is an isomorphism because the order of the group matches the degree of the extension. □

Example 4.6: The Case $n = 5$ (Cyclic Group)

Let ζ_5 be a primitive fifth root of unity. The cyclotomic polynomial is

$$\Phi_5(x) = \frac{x^5 - 1}{x - 1} = x^4 + x^3 + x^2 + x + 1.$$

By Theorem 4.4, $\Phi_5(x)$ is irreducible over $\mathbb{Q}$, so $[\mathbb{Q}(\zeta_5) : \mathbb{Q}] = \deg(\Phi_5) = 4$. The Galois group is

$$\mathrm{Gal}(\mathbb{Q}(\zeta_5)/\mathbb{Q}) \cong (\mathbb{Z}/5\mathbb{Z})^\times = \{1, 2, 3, 4\} \mod 5.$$

Since $2^1 = 2$, $2^2 = 4$, $2^3 \equiv 3$, and $2^4 \equiv 1$, the group is generated by the automorphism $\sigma_2 : \zeta_5 \mapsto \zeta_5^2$. Thus, the Galois group is cyclic of order 4:

$$\mathrm{Gal}(\mathbb{Q}(\zeta_5)/\mathbb{Q}) \cong \mathbb{Z}/4\mathbb{Z}.$$

Here is another example where the Galois group is not cyclic.

Example 4.7: The Case $n = 8$ (Klein-Four Group)

The primitive eighth roots of unity are ζ_8^k for $k = \{1, 3, 5, 7\}$. The eighth cyclotomic polynomial is

$$\Phi_8(x) = (x - \zeta_8)(x - \zeta_8^3)(x - \zeta_8^5)(x - \zeta_8^7) = x^4 + 1.$$

By Theorem 4.4, $x^4 + 1$ is irreducible over $\mathbb{Q}$, so

$$[\mathbb{Q}(\zeta_8) : \mathbb{Q}] = \varphi(8) = 4.$$

The Galois group is

$$\mathrm{Gal}(\mathbb{Q}(\zeta_8)/\mathbb{Q}) \cong (\mathbb{Z}/8\mathbb{Z})^\times = \{1, 3, 5, 7\} \mod 8.$$

Note that $3^2 = 9 \equiv 1, 5^2 = 25 \equiv 1$, and $7^2 = 49 \equiv 1$. Since every non-identity element has order 2, the group is not cyclic. Therefore, it is the Klein Four-group:

$$(\mathbb{Z}/8\mathbb{Z})^\times \cong (\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z}).$$

The study of cyclotomic extensions illustrates the central philosophy of Galois theory: the algebraic structure of a field extension is reflected in the symmetries of its roots. In the cyclotomic setting, these symmetries admit an especially explicit description. Every automorphism of $\mathbb{Q}(\zeta_n)$ is determined by its action on a primitive nth root of unity and this action corresponds naturally to multiplication by an element of $(\mathbb{Z}/n\mathbb{Z})^\times$. Consequently, the arithmetic structure of integers modulo n completely determines the structure of the associated Galois group. Cyclotomic fields therefore provide one of the clearest and most concrete examples of the deep relationship between field theory, group theory, and number theory, and they remain central objects in modern algebra.

5 Closing Remarks

A Bridge Between Algebra and Arithmetic

Cyclotomic fields occupy a central position in algebra and number theory because they provide an explicit, computable family of Galois extensions. While many splitting fields have elusive automorphism group, cyclotomic extensions are completely described by elementary arithmetic data. The isomorphism

$$\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$$

demonstrates how complex questions of field automorphisms translate directly into the language of modular arithmetic.

Geometry and the Gauss-Wantzel Theorem

Historically, the study of roots of unity resolved ancient questions regarding geometric constructibility. Since the vertices of a regular n-gon are defined by the nth roots of unity, constructing a polygon with straightedge and compass is equivalent to analyzing the field generated by ζ_n.

Theorem 5.1: Gauss-Wantzel's Theorem

A regular n-gon is constructible if and only if

$$n = 2^k p_1 \cdots p_r,$$

where the p_i are distinct Fermat primes.

This was first introduced by Carl Gauss (1777 - 1855) and later generalized and proven by Pierre Wantzel (1814 - 1848). This theorem occurs because constructibility requires a tower of quadratic extensions, meaning the degree $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$ must be a power of 2.

Foundations of Kummer Theory

The work of Ernst Kummer (1810 - 1893) generalized these ideas into what we now call Kummer extensions. By adjoining nth roots of elements (extensions of the form $K(\sqrt[n]{a})$), Kummer found that their structure is most "well-behaved" when the base field already contains the nth roots of unity. Originally motivated by attempts to prove Fermat's Last Theorem, his work established cyclotomic fields as the natural setting for understanding radical extensions.

The Kronecker–Weber Theorem

Cyclotomic fields are the cornerstone of class field theory, which studies abelian extensions of number fields. The Kronecker–Weber Theorem provides a profound classification of abelian extensions.

Theorem 5.2: Kronecker–Weber Theorem

Every finite abelian extension of the rational numbers $\mathbb{Q}$ is contained within some cyclotomic field.

Essentially, all abelian symmetry over the rationals arises from adjoining roots of unity.

Legacy in Modern Mathematics

Beyond pure number theory, roots of unity are indispensable across the mathematical landscape:

- **Representation Theory:** Cyclotomic extensions provide the character values and splitting fields necessary to decompose group algebras, ensuring that representations can be fully realized over specific abelian extensions of $\mathbb{Q}$.
- **Harmonic Analysis:** Through roots of unity, they give the discrete arithmetic structure necessary to define the Fourier transform on finite abelian groups, linking group characters and classical periodic analysis.
- **p-adic Number Theory:** Cyclotomic extensions serve as the local foundation for **Lubins-Tate theory**, where the p-adic completion of cyclotomic fields allows for the explicit construction of maximal abelian extensions of $\mathbb{Q}$.

- **Iwasawa Theory:** By studying the growth of ideal class groups within the $\mathbb{Z}_p$-extension of cyclotomic fields, Iwasawa theory links the purely algebraic properties of these extensions to the analytic values of p-adic L-functions.
- **Arithmetic Geometry:** Cyclotomic extensions describe the torsion points of the multiplicative group $\mathbb{G}_m$, providing the simplest examples of how Galois representations on the co-homology of algebraic varieties (like curves or abelian varieties) encode deep arithmetic information.

Ultimately, cyclotomic extensions transcend their algebraic origins to provide a universal language for the interplay between symmetry and arithmetic.

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